Project Notes: Post-Quantum Secure Communication for Autonomous Vehicle Platoons

Topic 1: Base Paper Summary vs. Capstone Project Plan

Summary of the Base Paper The base paper addresses the threat of future quantum computers breaking current cryptographic systems 1. To solve this, the authors upgraded Transport Layer Security (TLS) using two quantum-proof technologies:

1. **Post-Quantum Cryptography (PQC):** Complex mathematical algorithms (CRYSTALS-Kyber and CRYSTALS-Dilithium) 2.
2. **Quantum Key Distribution (QKD):** Physics-based key sharing using light particles over optical fibers 2, 3.They tested a **hybrid approach** combining both 1. Using PQC alone sped up the handshake by 9%, but combining PQC and QKD increased communication time by approximately 117% 1.

Differences from Your Project Plan

- **Application Domain:** The base paper focuses on static infrastructure connected by optical fibers 2, 3, while your project targets moving autonomous vehicle platoons exchanging Over-The-Air (OTA) updates and V2V communications 4-6.
- **Cryptographic Approach:** The base paper uses a hybrid QKD-PQC model 1, whereas your project proposes a pure post-quantum architecture, eliminating QKD hardware 5, 6.
- **Hardware:** The base paper uses specialized commercial QKD hardware 3. Your plan relies on standard vehicular hardware and software libraries like liboqs 5, 7.
- **Performance Priorities:** The base paper prioritizes absolute security at the cost of high latency (117% overhead) 1, 8, whereas your project prioritizes ultra-low latency, as platoons must exchange 10 to 50 critical messages per second 9.

Topic 2: Which Approach is Better?

When Your Project Plan is Better (Vehicular Networks) Your pure PQC architecture is the right choice for moving vehicles. QKD requires stationary optical fibers, which are impossible for highway driving 3, 8. Furthermore, autonomous platoons need to exchange 10 to 50 messages per second 9; the massive 117% communication overhead introduced by QKD would make real-time braking or steering impossible 1. Your literature survey rightly notes that the base paper's method has higher overhead and is not designed for vehicular environments 10.

When the Base Paper is Better (Static Infrastructure) The base paper's hybrid architecture is better for data centers or government networks where absolute, long-term security is paramount 11, 12. By combining PQC (math-based) and QKD (physics-based), it provides a "belt and suspenders" approach guaranteeing data is safe even if a flaw is discovered in the mathematical algorithms later 12.

Topic 3: Your System Summary & Efficiency

System Summary Your methodology breaks down into three core operations:

1. **OTA Server:** Sends firmware updates digitally signed using Dilithium, which vehicles verify via the liboqs library 5.
2. **Vehicle Node:** Uses Kyber for secure key exchange, combined with HKDF to derive a session key 5.
3. **Secure V2V Communication:** Platoon messages are encrypted using the fast AES-256-GCM standard, ensuring ultra-low latency while preventing tampering 5.

Ways to Make Your System More Efficient

1. **Optimize Parameter Sets:** Use lighter algorithm sizes like Kyber512 and Dilithium1 for real-time messages, as they offer the fastest performance 13.
2. **Manage MTU Fragmentation:** Large post-quantum keys cause network fragmentation because standard networks have a Maximum Transmission Unit (MTU) of 1500 bytes 11. Optimize certificates to reduce network traffic.
3. **Leverage Hardware Acceleration:** Instead of running heavy lattice-based math in Python, offload the processing to embedded automotive systems (like the ESP32), as noted in your literature survey 14.

Topic 4: Speed of Your Model

Your model is built to sustain 10 to 50 critical messages per second 9. It achieves this speed in two phases:

- **Connection Setup:** By using lightweight PQC algorithms (Kyber512 and Dilithium1), your secure connection handshake can be 10 to 11 milliseconds faster than traditional classical cryptography, yielding a 9% speed improvement 1, 13.
- **Message Encryption:** Once the session key is established, the continuous data stream is encrypted using AES-256-GCM, an industry-standard algorithm designed for high-speed transmission 5. By explicitly omitting QKD, you avoid the 117% latency delay 8.

Topic 5: Drawbacks of Your System

1. **Network Overhead:** Post-quantum keys are massive. A Dilithium3 certificate is ~4460 bytes, compared to a standard RSA certificate's 784 bytes 15. Because standard MTU is 1500 bytes 11, these keys must be broken into multiple packets, increasing latency.
2. **Algorithmic Vulnerability:** Relying purely on Kyber and Dilithium means your entire system is vulnerable if mathematicians eventually discover a way to break these specific lattice-based algorithms 12.
3. **Computational Latency in Software:** Processing complex cryptographic math in Python using liboqs creates computational overhead that may struggle to meet split-second vehicular decision-making requirements 7, 16.
4. **Lack of Real-World Validation:** Because city-scale scalability and real-world road testing are "Out of Scope" 5, your protocol's resilience under actual 5G-V2X highway interference remains unproven.

Topic 6: Optimal Solutions to Overcome Drawbacks

1. **Use Lighter Parameters:** To solve network overhead and packet fragmentation, exclusively use the lightest parameter sets (Kyber512 and Dilithium1) for high-speed

V2V communication. This reduces key sizes and generates a key exchange 11 milliseconds faster than classical models 13.

2. **Adopt a Classical-PQC Hybrid:** To mitigate algorithmic vulnerability, implement a hybrid key exchange combining Kyber with a pre-quantum algorithm like Elliptic Curve Diffie-Hellman (ECDHE) 12. This guarantees security as long as at least one algorithm remains unbroken.
3. **Hardware Acceleration:** To solve software latency, transition from Python to hardware-accelerated implementations on embedded systems, drawing on your literature survey references like the ESP32 implementations 7, 14.
4. **Network Simulation:** To compensate for the lack of real-world road testing, use advanced network simulators (like OMNeT++ or NS-3) to mimic vehicular traffic scenarios and 5G-V2X signal interference, as referenced in your survey of Vinel (2012) 4.